SLIDE 1/8 • SYSTEM DESIGN BREAKDOWN

Toy RAG vs Enterprise Production RAG

Why 80% of online LangChain tutorials break in production — and the 5 architectural fixes every backend engineer must know.

The Illusion

10-line Python scripts make RAG look solved in tutorials.

The Reality

In real enterprises: noisy PDFs, 8-second latency spikes, and silent hallucination.

The Goal

Convert naive vector lookups into an industrial-grade, observable pipeline.

SLIDE 2/8 • STAGE 1: INGESTION & CHUNKING

Naive Character Chunking vs Structural Parsing

How you split text determines 70% of your retrieval precision.

'L Toy Demo Approach

Fixed character chunking (e.g. 500 chars with 50 overlap). Splits financial tables and code blocks mid-sentence.

' Production Approach

Layout-aware semantic parsing (Docling / Unstructured). Extracts tables as Markdown and preserves hierarchical headers (H1/H2).

&¡ Production Metric

Reduces cross-chunk retrieval dropouts by 42%.

SLIDE 3/8 • STAGE 2: RETRIEVAL & SEARCH

Pure Dense Vector vs Hybrid Search (BM25 + Vector)

Why dense embeddings alone fail on exact IDs, part numbers, and error codes.

'L Toy Demo Approach

100% Cosine Similarity on dense embeddings (text-embedding-3). Fails on exact queries like 'Error code 0x80070057'.

' Production Approach

Hybrid Search via Reciprocal Rank Fusion (RRF): Combines BM25 Sparse Keyword Search (for exact IDs) with Dense Vectors (for semantic intent).

&¡ Production Fix

Elasticsearch / Qdrant hybrid index with alpha weighting.

SLIDE 4/8 • STAGE 3: PRECISION FILTERING

Blind Context Stuffing vs Cross-Encoder Reranking

Don't send 20 noisy chunks to your LLM — rerank them first.

'L Toy Demo Approach

Fetch top 15 chunks and paste them all into the prompt context window. Drives up token costs and causes 'Lost in the Middle' hallucinations.

' Production Approach

Two-stage retrieval: Retrieve top 40 candidates fast, then apply a Cross-Encoder (Cohere Rerank / BGE-Reranker) to pass only top 4 pristine chunks.

&¡ Impact

Cuts LLM inference token costs by 65% while increasing answer precision.

SLIDE 5/8 • STAGE 4: SERVING & CACHING

Synchronous Blocking vs Semantic KV Caching

How enterprise systems achieve sub-200ms latency on repeated queries.

'L Toy Demo Approach

Every user query triggers a cold vector search + full LLM generation ($0.03 per call, 2.8s latency).

' Production Approach

Semantic Caching (GPTCache / Redis Vector): Matches semantically identical queries (cosine > 0.96) and returns cached answers in <40ms.

&¡ Production Metric

Saves up to 40% of OpenAI/Claude API billing in enterprise apps.

SLIDE 6/8 • STAGE 5: AUTOMATED EVALUATION

Vibes-Based Testing vs Automated CI/CD Ragas Triad

How to catch regressions before your customers find them.

'L Toy Demo Approach

The developer types 3 test questions into a Streamlit UI, says 'looks good to me', and deploys.

' Production Approach

Automated RAG Triad in GitHub Actions with Ragas / TruLens: Faithfulness Score (>0.90), Answer Relevance (>0.88), and Context Precision.

&¡ CI/CD Rule

Fails the build if faithfulness drops below 0.92 on benchmark suite.

SLIDE 7/8 • THE PRODUCTION ARCHITECTURE BLUEPRINT

The Complete Enterprise RAG Stack

The 6 core layers of a battle-tested enterprise retrieval system.

1. Docling Parser

Structural & table-aware chunking

2. Hybrid Index

BM25 + Dense Vector (Qdrant/Pinecone)

3. Cohere Reranker

Top 40 -> Top 4 precision filter

4. Redis Semantic Cache

Sub-50ms instant response cache

5. Guardrails & Ragas

Prompt injection & faithfulness eval

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